

An Ontology-Guided Approach for Enhanced Documentation of Cave Microbial Culture Collections

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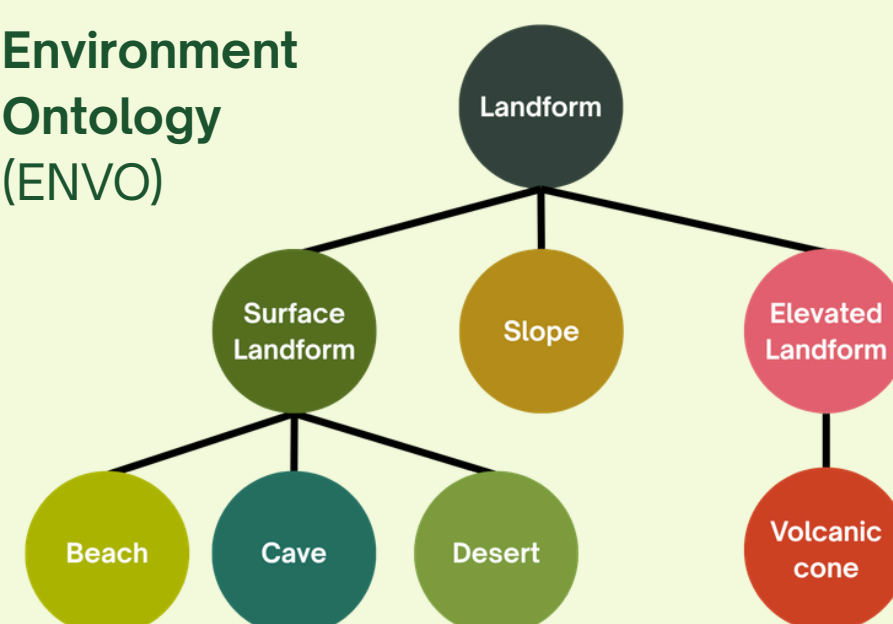
1 Introduction

Free-text annotations in cave culture collection information systems lead to **inconsistent** descriptions of **identical** concepts, making cave microbial data difficult to interpret and compare.

low temperature === cold ?

Ontologies bring a unique solution by providing **standardized terms** with defined relationships, promoting accuracy in data entry.

Environment Ontology (ENVO)



2 Objectives

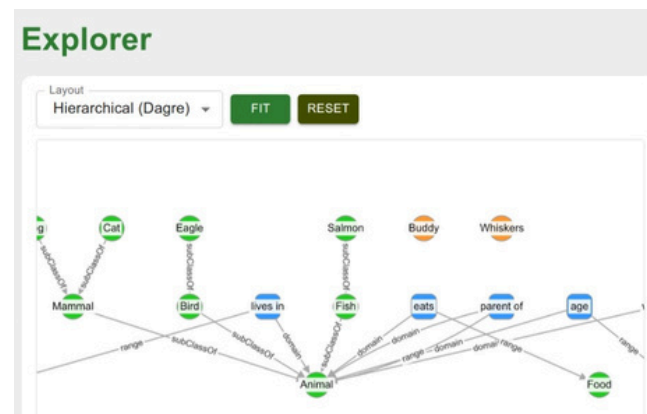
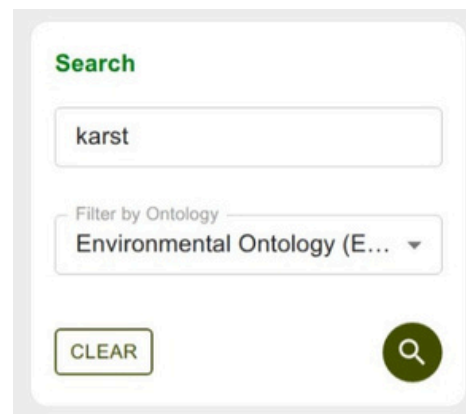
Escherichia coli
<http://purl.bioontology.org/ontology/NCBITAXON/562>

Balantoides coli
<http://purl.bioontology.org/ontology/NCBITAXON/71585>

Entamoeba coli
<http://purl.bioontology.org/ontology/NCBITAXON/110766>

Aedoeadaptatus coli
<http://purl.bioontology.org/ontology/NCBITAXON/2058292>

- to integrate **ontology-based suggestions** on free-text annotations to support more consistent data entry;



- to develop an **Ontology Lookup Service (OLS)** to help users explore relevant ontology classes and an **Ontology Explorer Service (OES)** for examining their relationships; and

System Usability Scale Questionnaire	Strongly Disagree	Strongly Agree
1. I think that I would like to use this product frequently.	1 2 3 4 5	
2. I found the product unnecessarily complex.	1 2 3 4 5	
3. I thought the product was easy to use.	1 2 3 4 5	
4. I think that I would need the support of a technical person to be able to use this product.	1 2 3 4 5	
5. I found the various functions in the product were well integrated.	1 2 3 4 5	
6. I thought there was too much inconsistency in this product.	1 2 3 4 5	

- to evaluate the system using the **System Usability Scale (SUS)**.

3 Methodology

The Caventology web application was developed with:

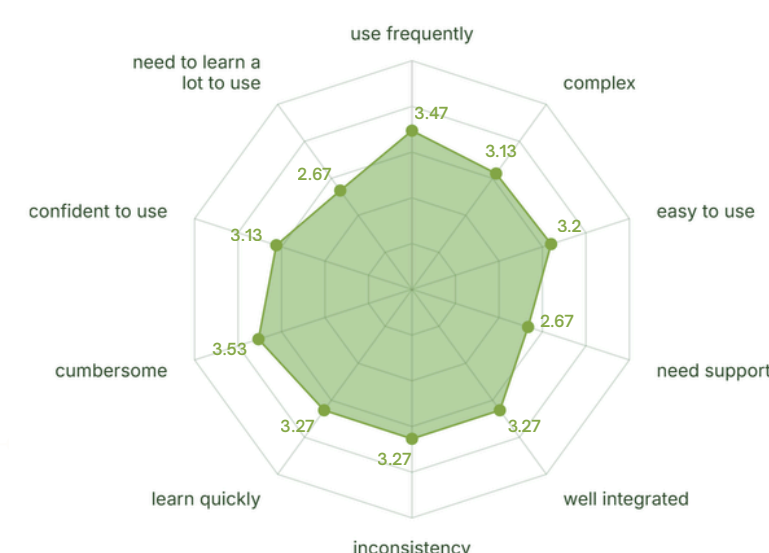
- React.js** + **Material UI (MUI)** for the user interface,
- Django** for the server, and **MySQL** for database management.
- The OES uses **N3.js** for parsing RDF ontology files and **Cytoscape.js** for visualization.
- The OLS and ontology-based suggestions are powered by the **BioPortal Search API**.



4 Results

- 15** respondents evaluated the system using **SUS**.
- Researchers appreciate the **tight integration** of the application's features.

- The system received a mean SUS score of **79**, indicating **good** overall usability.



5 Conclusion

By integrating ontology-based suggestions along with OLS and OES tools in the cave culture collection information system, **Caventology** enhances the consistency and semantic quality of cave microbial documentation.



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